

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

● ANSWER KEY

Probability and conditional probability

03

3 answers · Rationale-rich · Teach from doc

Q1**B**

B. $\frac{415}{799}$

Choice B is correct. If one of the teens surveyed is selected at random, the probability that the teen talks on a cell phone daily is equal to the quotient of the total number of teens who reported that they talk on a cell phone daily, 415, and the total number of teens surveyed, 799.

Therefore, this probability is equal to $\frac{415}{799}$.

Choice A is incorrect. This fraction represents the probability of selecting at random any one of the 799 teens surveyed. Choice C is incorrect and may result from conceptual errors. Choice D is incorrect. This fraction represents the probability of selecting at random one of the 799 teens surveyed who doesn't talk on a cell phone daily.

Q2**A**

A. $\frac{1}{14}$

Choice A is correct. The total number of possible outcomes for rolling a fair 14-sided die is 14. The number of possible outcomes for rolling a 2 is 1. The probability of rolling a 2 is the number of possible outcomes for rolling a 2 divided by the total number of possible outcomes, or $\frac{1}{14}$.

Choice B is incorrect. This is the probability of rolling a number no greater than 2.

Choice C is incorrect. This is the probability of rolling a number greater than 2.

Choice D is incorrect. This is the probability of rolling a number other than 2.

Q3**B****B.** $\frac{9}{50}$

Choice B is correct. If one of these children is selected at random, the probability of selecting a child who chose a vegetarian sandwich is equal to the number of children who chose a vegetarian sandwich divided by the total number of children. According to the table, there are a total of 50 children, and 9 of these children chose a vegetarian sandwich. Therefore, the probability of selecting a child who chose a vegetarian sandwich is $\frac{9}{50}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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